

ZEYU (ALBAN) LI, Ph.D.

Cross-Disciplinary Scientist-Engineer | Agentic AI, Evaluation & Scientific R&D Systems

Building Reliable AI Workflows Grounded in Experimental Judgment

Ithaca, NY | 626-826-5327 | zl788@cornell.edu | zeyuli.net/ai | linkedin.com/in/albanli | github.com/AlbanLi0314

SUMMARY

Cornell Ph.D. scientist-engineer building agentic AI, retrieval, evaluation, and data systems for scientific R&D. Scientific background spans materials, biological, environmental, and molecular systems, alongside undergraduate training in chemistry and computer science and hands-on experience in automation hardware and data-intensive experimental workflows. Builds AI systems that turn complex scientific work into structured workflows, traceable outputs, and evidence-aware evaluation layers, using domain judgment to identify when model outputs conflict with experimental constraints or physical evidence. AI work includes Research Harness, a 13-skill agentic framework for wet-lab research judgment; RAG-based researcher and literature discovery; and real-time multilingual voice agents. Research record spans environmental monitoring technologies, self-healing structural materials, continuous-flow biomanufacturing platforms, and patterned polymer surface microfabrication, alongside an outstanding teaching award for AI curriculum design, a first-author ICML AI for Science workshop paper, 5 publications, and 2 granted patents.

SKILLS

LLM & Agentic AI Systems: OpenAI, Claude, Gemini, APIs; prompt and context engineering; agentic workflow design; agent and multi-agent orchestration; tool/function calling, routing, structured outputs, and schema-bounded tool contracts; RAG over external knowledge; AI-assisted code generation and development workflows; LLM evaluation and benchmarking; hallucination detection, LLM-as-judge, guardrail and safety-layer design, human-in-the-loop validation; streaming ASR and real-time voice interfaces (WebSocket, Web Audio API); AI-assisted development tools including Claude Code, Cursor, and Copilot

Machine Learning & Retrieval: PyTorch, TensorFlow, XGBoost; semantic search, hybrid retrieval (vector + sparse), query expansion, retrieval evaluation; FAISS, BM25, Reciprocal Rank Fusion, SPECTER2 embeddings

Scientific Data Engineering & Visualization: Python data stack (pandas, NumPy, GeoPandas, Matplotlib/Seaborn); ETL pipelines, schema validation, multi-source data integration, reproducible scientific data workflows; geospatial analysis, time-series analysis, scientific visualization

Full-Stack & Deployment: React 18/19, Electron, Streamlit; FastAPI, REST APIs; GCP Cloud Run, Firebase (Auth, Firestore, Realtime Database), Vercel; Docker; Git/GitHub

Experimental Automation Hardware & Scientific Instrumentation: microfluidics; experimental automation; device fabrication and processing (3D printing: DIW/SLA/FDM, electrospinning, photolithography, thermal imprinting, cleanroom protocols); materials characterization (SEM/EDS, rheometry, DLS, tensile testing, contact-angle measurement); biological assays (qPCR/PCR, in vitro transcription, gel electrophoresis, UV-Vis spectroscopy, fluorometry); design and visualization software (CAD/SketchUp, Blender, KeyShot, Adobe Illustrator/Photoshop)

Languages: Python, Java; English, Mandarin, Cantonese

AI-FOR-SCIENCE SYSTEMS & APPLIED AI AGENTS

Agentic Scientific Workflows, Retrieval & Evaluation Systems

Research Harness — Agentic Workflow and Evaluation System for Wet-Lab Research Judgment

Agentic AI, Scientific Judgment Capture, LLM Evaluation, Guardrails, Human-in-the-Loop, Traceability

- **Built a 13-skill agentic framework for LLM-assisted wet-lab research**, translating tacit experimental judgment into reusable scientific workflows with structured pre-output clarification, PASS/FAIL constraint checks, evidence-aware guardrails, and multi-agent review before final outputs.
- **Operated and iterated the framework across 130+ AI research sessions over the lifecycle of a wet-lab research project**, producing 2,269 indexed artifacts across 167 directories with bidirectional traceability from written outputs back to source data, claims, and researcher-defined constraints.
- **Encoded 33 researcher-defined rules into sessions to diagnose recurring AI failure modes**, enforce claim-evidence discipline, and improve alignment between model outputs and human wet-lab research judgment.
- **First-author paper** in ICML Workshop on AI for Science, 2026 | GitHub: github.com/AlbanLi0314/research-harness-skills

Resonance — Paper-Grounded Retrieval and Evaluation System for Researcher Discovery

Gemini 3 Hackathon | Deployed on Streamlit | Gemini API, SPECTER2, FAISS, BM25, RAG | Retrieval Evaluation

- **Built and deployed a paper-grounded AI application that helps researchers identify relevant institutional experts for scientific discussion:** users upload a paper, clarify their research goal through multi-turn LLM dialogue, and receive ranked researcher matches with AI-generated relevance explanations.
- **Engineered a hybrid retrieval pipeline over 230 researchers**, combining LLM-driven query expansion, SPECTER2 dense retrieval via FAISS, BM25 sparse retrieval, and Reciprocal Rank Fusion to connect paper-level scientific context with researcher expertise.
- **Built a 45-query gold evaluation set** with P@1, nDCG@5, and MRR metrics to measure retrieval quality, diagnose ranking failures, and iterate toward more reliable researcher-discovery results.

Scientific Data & Experimental Workflow Infrastructure

Automated Field Campaign Data Pipeline — Reproducible Scientific Data Engineering for a DoD-Funded Environmental Monitoring

Python, Pandas, GeoPandas, Seaborn | ETL, Geospatial Analytics, Data Validation, Scientific Visualization

- **Built a reproducible Python data pipeline for a DoD-funded lake-scale field monitoring study**, integrating ~1,000 molecular measurement records and ~1,400 GPS drifter coordinates from 278 sampling stations across an 11 sq. km study area into validated, structured outputs for downstream hydrodynamic model benchmarking.
- **Automated data validation, temporal synchronization, coordinate standardization, and multi-table output generation;** developed ~20 schema-enforced Python scripts for zero-touch regeneration of publication-ready geospatial maps and statistical plots from raw field data.

Automated Inline Chip Reactor — Continuous-Flow Platform for Perpetual Biomanufacturing

Microfluidic, Cartridge-Based Reactor, Inline Reactor, Bioprocessing

- **Designed and assembled modular continuous-flow chip reactors** integrating DNA-immobilized NucleoMesh cartridges, PDMS spacers, acrylic plates, tubing, syringe-pump flow control, and temperature-controlled operation for continuous RNA/DNA production.
- **Built and optimized the reactor hardware workflow** across flow rate, enzyme delivery, temperature, reagent conditions, and cartridge operation, translating a material-based nucleic-acid production concept into a controllable microfluidic biomanufacturing platform.

Automated Environment Simulation Chamber — Environmental Simulation Platform for Materials Testing

Environmental Simulation, Experimental Automation, Droplet Control, Electrostatic Surface Charging, High-Speed Monitoring

- **Designed and fabricated an automated environmental simulation chamber** for controlled droplet-impact testing, integrating temperature and humidity control, supercooled droplet generation, adjustable droplet parameters (angle, speed, frequency, size), sample-stage cooling and rotation, control panels, and high-speed monitoring.
- **Integrated an electrostatic surface-charging module** with a negative ion generator, high-voltage power supply, grounded shield, and sample stage to simulate charged icing environments and quantify voltage-dependent droplet contact time, expansion ratio, and rebound behavior across superhydrophobic surfaces.

Real-Time Voice & Multimodal AI Agents

Kinapse — Voice-First AI Workflow Coach for Prompting and App Context

Gemini Live Agent Hackathon | Real-Time Voice Agent | Electron + FastAPI | GCP Cloud Run | Multimodal Context | RAG Memory

- **Built and shipped a macOS voice agent that helps users improve their use of AI tools through real-time prompt coaching:** users speak naturally, compare their original input with an optimized prompt, apply the result directly within the active application, and track prompt-quality progression across sessions.
- **Engineered a real-time multimodal agent stack with Gemini Live streaming ASR**, sub-second interaction latency, an Electron client, and a Python/FastAPI backend deployed on GCP Cloud Run, combining voice, screen context, active-window state, and a two-layer RAG memory system for personalized prompt coaching across sessions.

ER Companion — Multilingual Voice Agent for Structured Emergency-Department Intake

Winning Project of Cornell AI Health Hackathon | Gemini Live API, WebSocket Streaming, Function Calling | Multilingual NLP

- **Built a patient-facing multilingual voice agent for emergency-department intake**, helping non-English-speaking patients describe symptoms in real time while structuring the conversation through OPQRST-guided LLM function calls and delivering parsed symptom histories to a nurse-facing dashboard.
- **Designed a safety and escalation layer** combining instant red-flag alerts, 150+ multilingual regex patterns across 4 condition categories, Gemini function-call severity screening with ESI-2 threshold logic, special-population adjustments, and automated 7-minute pain-score check-ins to support safer clinical handoff. | GitHub: github.com/AlbanLi0314/er-companion

SCIENTIFIC R&D EXPERIENCE

Graduate Research Assistant

Feb 2021 – May 2026

DNA Materials Lab, Advisor: Prof. Dan Luo | Cornell University, Ithaca, NY

Environmental Monitoring Nanobiotechnology — DNA-Barcoded PLGA nanoparticle Tracers for Lake-Scale Monitoring

Environmental Monitoring | Molecular Detection | Polymer Engineering

- **Designed, fabricated, and deployed DNA-barcoded biodegradable PLGA nanoparticle tracers** in a DoD SERDP-funded lake-scale monitoring study, coordinating a 33-hour, 11 sq. km field campaign across 4 disciplines and 7 research groups.
- **Developed the experimental and analytical workflow** for tracer recovery and TaqMan qPCR detection, enabling quantitative tracking over 7 km from milligram-scale DNA inputs under contamination-controlled conditions.
- **First-author paper** in *Environmental Science & Technology* (IF = 11.3), 2025.

Self-Healing Structural Materials — Biomass DNA Hydrogels for Multi-Mode Crosslinking and Structural Repair

Biomaterials | Self-Healing Materials | Computational Design

- **Developed a biomass-DNA hydrogel platform** using thermal, photochemical, and metal-ion crosslinking, spanning reversible, photo-cleavable, and irreversible network behaviors from a single low-cost DNA feedstock.
- **Established material processing and performance benchmarks** through rheology, swelling/retention assays, diffusion-limited gelation studies, tensile testing, and extrusion-based 3D printing.
- **Developed DNA hydrogel systems for architectural ceramic tile prototypes**, enabling proof-of-concept self-healing structural repair through separate internal channels for DNA and $AlCl_3$ solutions that form localized gel at crack sites.
- **First-author paper** in *ACADIA, computational design and digital fabrication premier conference*, 2024.

Continuous-Flow Perpetual Biomanufacturing — NucleoMesh Chip Reactor Platform for Long-Duration DNA/RNA Production

Biomanufacturing | Microfluidic Reactors | Continuous-Flow Processing

- **Developed a continuous-flow nucleic acid production platform** using DNA-immobilized NucleoMesh cartridges and syringe-pump-operated chip reactors for multi-day enzymatic RNA/DNA production.
- **Optimized and validated NucleoMesh material performance** through hydrogel formulation, template immobilization, SEM, rheology, fluorescence imaging, gel electrophoresis, and template-loading quantification.
- **Second-author paper** in *Nature Sustainability*, in peer review, 2026.

Low-Cost DNA/RNA Extraction System — Pod-Based In-line Platform for DNA/RNA Capture and Processing

Biochemical Engineering | Nucleic Acid Extraction | Process Engineering

- **Designed an espresso-machine-inspired nucleic acid extraction platform** using a pod-based continuous-flow format to target automatic, high-capacity, and high-throughput processing of nucleic acid samples.
- **Established perfluorocarbons as high-capacity DNA capture media, with perfluorodecalin reaching 41.73 $\mu\text{g}/\mu\text{L}$ adsorption capacity, roughly 800 $\times$ higher than silica-based methods and translating to an estimated 209 mg DNA loading per pod.**
- **Reduced projected consumables cost** by approximately 91% compared with conventional ethanol precipitation while maintaining over 90% DNA capture efficiency.

Senior Research Assistant & Undergraduate Research Assistant

Jun 2017 – Jan 2021

Microfabrication & Surface Materials Lab, Supervisor: Prof. Kangning Ren | Hong Kong Baptist University

Patterned Polymer Surface Microfabrication — Durable All-Perfluoropolymer Superhydrophobic Materials for Icing Control

Surface Engineering | Materials Science | Durable Superwettability

- **Engineered and validated all-perfluoropolymer monolithic superhydrophobic surfaces** using high-temperature soft imprinting and thermal fusion to combine controlled 3D microtopology with perfluoropolymer chemistry, maintaining topology-specific superwettability under abrasion, compression, corrosive vapors, UV aging, thermal treatment, and 100,000+ folding cycles
- **Third-author paper** in *The Innovation* (IF = 33.2), 2023; **co-inventor**, U.S. Patent 11,839,998; recipient, Best Undergraduate Thesis Award.

Research Exchange Trainee

Jun 2018 - Aug 2018

Molecular Basis of Disease Program | Georgia State University, Atlanta, GA

- **Characterized transcription factor-DNA binding kinetics** (PU.1) via LSPR assays, quantifying sequence-specific binding affinities in real time. Best Poster Presentation Award.

TEACHING & AI CURRICULUM

Co-Instructor & Teaching Assistant, AI Engineering Curriculum Design

Feb 2021 - May 2026

Cornell University, Department of Biological and Environmental Engineering

- **Designed and delivered a 6-lecture AI tooling module for engineering students**, covering LLM fundamentals, prompt and context engineering, agentic workflow design, and AI-assisted data processing.
- **Led a 40-student biomaterials AI hackathon** with structured deliverables and oral defense, guiding students to translate AI methods into applied scientific workflow designs.
- **Guest lectured on AI-assisted engineering workflows** in 2 additional undergraduate courses, introducing transformer attention, context engineering, the 40-20-40 rule, and practical agent design patterns.
- **Supported teaching across 7 engineering and bioengineering courses** spanning nucleic acid engineering, biomaterials, molecular and cellular bioengineering, watershed engineering, and engineering professionalism; recipient of the Cornell CALS Outstanding Teaching Assistant Award.
- **Supervised 3 undergraduate researchers** on wet-lab protocols, experimental troubleshooting, and data analysis workflows.

EDUCATION

Cornell University

Ithaca, NY

Biological and Environmental Engineering | Concentration: Materials Science, Biological Engineering, Bioenvironmental Engineering.

Ph.D. | May 2026 | **M.S.** | Aug 2024 | **M.Eng.** | May 2020

Hong Kong Baptist University

Kowloon, Hong Kong

Chemistry (Major) | Computer Science (Minor)

B.Sc. (Hons) | Nov 2019

PUBLICATIONS

Li, Z., Liu, Z., Yang, X., Luo, D. "What the Records Don't Carry: A Position on Researcher-AI Co-Adaptation in Exploratory Laboratory Research." ICML Workshop on AI for Science, 2026.

Wang, D., **Li, Z.**, Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." Nature Sustainability, in peer review, 2026.

Li, Z., Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling." Environmental Science & Technology (IF = 11.3), 2025. [DOI](#)

He, C., **Li, Z. (co-first authors)**, Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles." ACADIA 2024. [DOI](#)

Li, W., Chan, C. W., **Li, Z.**, et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." The Innovation (IF = 33.2), 4(2), 100389, 2023. [DOI](#)

PATENTS

Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., **Li, Z.**, Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998 (granted 2023).

Li, Q., **Li, Z.**, & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (granted 2017).